

PROBABILITY AND STATISTICS

Lectures 3-4

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Outline I

- 1 Normal distribution
- 2 Joint Distribution
- 3 Covariance. Coefficient of correlation
 - Covariance and Correlation in R
- 4 Limit theorems
 - Markov's and Chebyshev's inequalities
 - 3σ 6σ rules
 - Law of large Numbers
- 5 Central Limit Theorem (CLT)

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https://www.youtube.com/watch?v=KkC_k7HVJyY

Example 3: Normal (Gaussian) Distribution

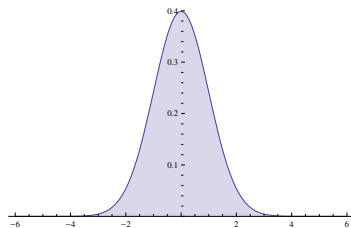
The most important distribution in statistics, characterized by its bell shape.

PDF

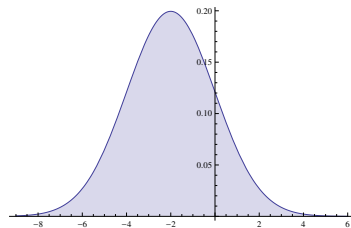
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ is the mean (location), σ is the standard deviation (scale).
- Examples: Distribution of household income, stock market returns (under certain assumptions), and measurement errors.
- $\xi \sim N(\mu, \sigma^2)$

Normal Distribution



$$\mu = 0, \sigma = 1$$

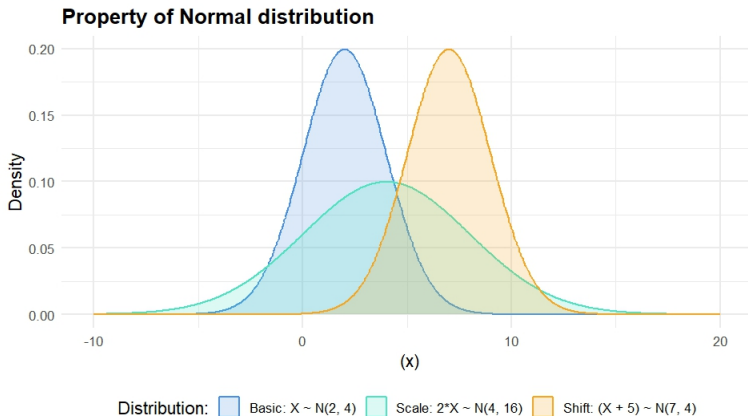


$$\mu = -2, \sigma = 2$$

PDF of normal distribution

Properties

- $\xi \sim N(\mu, \sigma^2) \Rightarrow E\xi = \mu, \text{Var}\xi = \sigma^2.$
- If $\xi \sim N(\mu, \sigma^2)$, then $\xi a + b \sim N(\mu a + b, a^2 \sigma^2).$



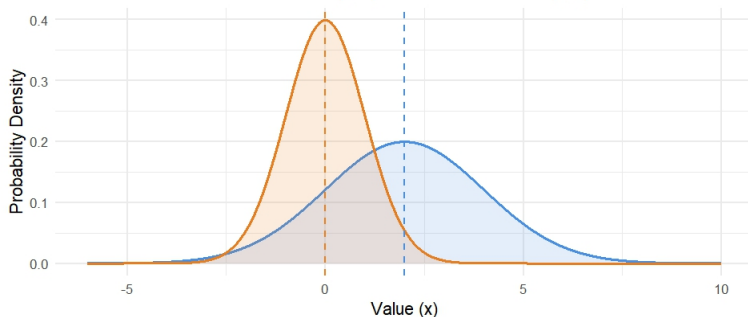
Z procedure

Remark

This property allows any normally distributed random variable to be reduced to the standard Gaussian distribution, specifically to $\mathcal{N}(0, 1)$. If $\xi \sim N(\mu, \sigma^2)$, to $z = \frac{\xi - \mu}{\sigma} \sim N(0, 1)$.

Theoretical Z-Score Transformation

Mathematical Shift and Scale from $N(2, 4)$ to Standard Normal $N(0, 1)$



Distribution: ■ Original Distribution: $X \sim N(2, 4)$ ■ Standardized Distribution: $Z \sim N(0, 1)$

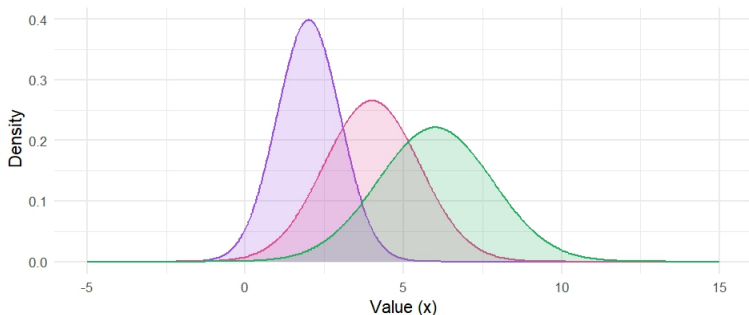
Additivity property

If ξ_i are independent rv $\xi_i \sim N(\mu_i, \sigma_i^2)$, $i = \overline{1, n}$, then

$$\sum_{i=1}^n \xi_i \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right).$$

Additivity Property of Normal Distributions

The theoretical sum of two independent normal variables is also normally distributed



Variable / Distribution: $X \sim N(2, 1)$ $Y \sim N(4, 2.25)$ Сума: $(X + Y) \sim N(6, 3.25)$

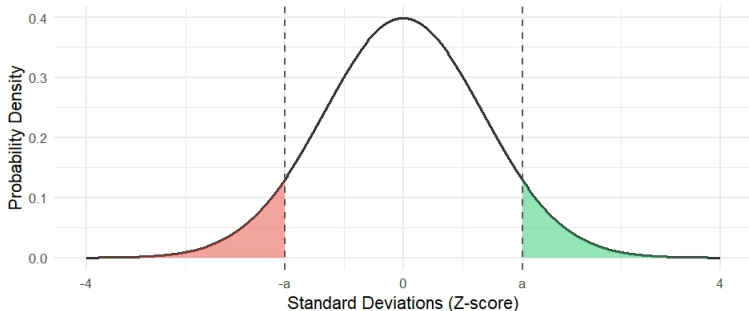
- If $z \sim N(0, 1)$, then

$$\Phi(-a) = 1 - \Phi(a),$$

where $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{t^2}{2}} dt$.

Symmetry of Normal Distribution Tails

Visual proof of the property: $\Phi(-a) = 1 - \Phi(a)$



Symmetrical Areas: ■ Left Tail: $\Phi(-a)$ ■ Right Tail: $1 - \Phi(a)$

Example

A beverage company fills aluminum cans with cola. The machine is calibrated so that the volume of liquid follows a normal distribution with a mean (μ) of 355 mL (the advertised volume) and a standard deviation (σ) of 1.5 mL.

The company's quality assurance policy states that any can containing less than 352 mL is considered “under-filled”(which risks legal issues for false advertising) and any can containing more than 358 mL is “over-filled” (which wastes product and money).

- What is a probability that can is under-filled? ($P(X < 352)$)
- What is a probability that can meets the company's quality standards? ($P(352 \leq X \leq 358)$)

Solution. Z-score approach

$$X \sim N(355, 1.5^2)$$

Convert the value X into its corresponding Z -score

$$\begin{aligned} P(X < 352) &= P\left(\frac{X - \mu}{\sigma} < \frac{352 - \mu}{\sigma}\right) = \\ &= P(Z < -2.00) = \Phi(-2.00) \approx 0.02275. \end{aligned}$$

Approximately **2.28%** of the cans will be under-filled.

Acceptable Quality

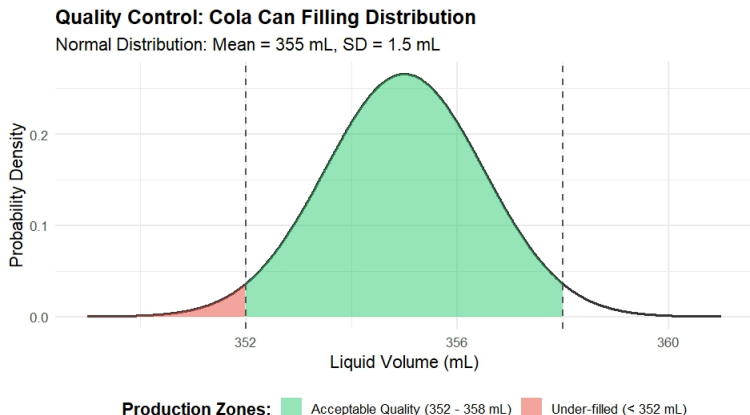
We want to find the probability that a can meets the quality standards (352 mL to 358 mL):

$$\begin{aligned}P(352 \leq X \leq 358) &= P\left(\frac{352 - 355}{1.5} \leq Z \leq \frac{358 - 355}{1.5}\right) \\&= P(-2.00 \leq Z \leq 2.00) = \Phi(2.00) - \Phi(-2.00) \\&= 0.97725 - 0.02275 = 0.9545\end{aligned}$$

Exactly **95.45%** of production meets the acceptable quality thresholds (conforming to the Empirical Rule for $\pm 2\sigma$).

$$P(X < 352) \quad - \quad pnorm(352, 355, 1.5)$$

$$P(352 < X < 358) \quad - \quad pnorm(352, 355, 1.5) - pnorm(358, 355, 1.5)$$

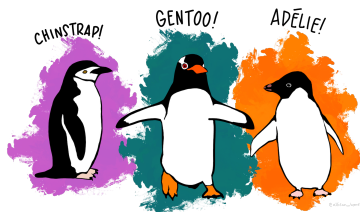


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Palmer Penguins

The Palmer Penguins dataset (packaged as **palmerpenguins** in R) is an dataset collected by Dr. Kristen Gorman, the Palmer Station, Antarctica LTER, it contains real-world data for 344 penguins across three distinct species and three islands in the Palmer Archipelago.

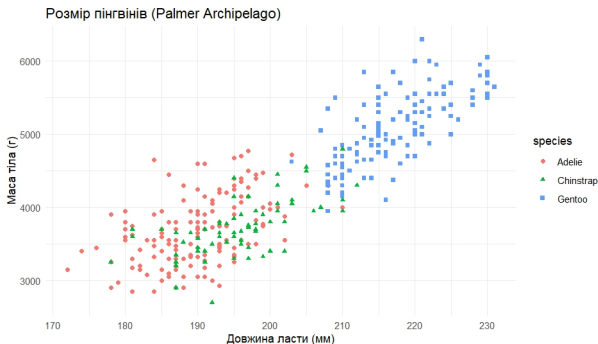


Starting with R

```
library(palmerpenguins)
library(ggplot2)

data("penguins")
head(penguins)
summary(penguins)
penguins_final <- na.omit(penguins)
ggplot(data = penguins, aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point(aes(color = species, shape = species))

table(penguins_final$species, penguins_final$island)
```



Discrete Case: Joint PMF

If X and Y are discrete random variables, their joint behavior is defined by the Joint Probability Mass Function (PMF):

$$p(x, y) = P(X = x, Y = y)$$

Properties of Discrete Joint PMFs

- 1 $0 \leq p(x, y) \leq 1$ for all possible outcomes (x, y) .
- 2 The sum over all possible pairs must equal 1:

$$\sum_x \sum_y p(x, y) = 1$$

Continuous Case: Joint PDF

If X and Y are continuous random variables, their joint behavior is modeled via a surface called the **Joint Probability Density Function (PDF)**: $f(x, y)$.

Probability is represented as the **volume** under the surface $f(x, y)$ over a region A in the XY -plane:

$$P((X, Y) \in A) = \iint_A f(x, y) dx dy$$

Properties of Continuous Joint PDFs

$$\sum \implies \int \quad \text{and} \quad \iint_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Marginal Distributions

If we know the joint distribution but want to find the individual distribution of just **one** variable, we calculate its **Marginal Distribution**. This process "sums out" or "integrates out" the other variable.

Discrete Case

$$p_X(x) = \sum_y p(x, y)$$

$$p_Y(y) = \sum_x p(x, y)$$

Continuous Case

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Statistical Independence

Two random variables X and Y are **statically independent** if the occurrence or value of one does not affect the probability distribution of the other.

Definition of Independence

X and Y are independent if and only if their joint distribution is the exact product of their individual marginal distributions for all values:

$$\text{Discrete: } p(x, y) = p_X(x) \cdot p_Y(y)$$

$$\text{Continuous: } f(x, y) = f_X(x) \cdot f_Y(y)$$

If this multiplicative rule fails at even a single point, the variables are **dependent**.

Example

Let X be the number of smartphones and Y be the number of smartwatches owned by a consumer in a tech study. Their joint PMF $p(x, y)$ is given by the table below:

$X \setminus Y$	$Y = 0$	$Y = 1$	$Y = 2$	Marginal $p_X(x)$
$X = 1$	0.10	0.40	0.10	0.60
$X = 2$	0.05	0.25	0.10	0.40
Marginal $p_Y(y)$	0.15	0.65	0.20	1.00

Calculations from the Table

- **Joint Probability:** The probability a person owns 2 phones and 1 watch is $p(2, 1) = 0.25$.
- **Marginal Probability:** The probability a person owns 1 phone regardless of watches is $p_X(1) = 0.10 + 0.40 + 0.10 = 0.60$.

Example

Using the previous smartphone (X) and smartwatch (Y) joint distribution table, let us mathematically test if the variables are statistically independent.

Testing the Independence Condition: $p(x, y) = p_X(x) \cdot p_Y(y)$

Let's test the specific point where $X = 1$ and $Y = 0$:

- From the joint cell: $p(1, 0) = 0.10$
- From the marginal totals: $p_X(1) = 0.60$ and $p_Y(0) = 0.15$

Multiplying the marginals gives:

$$p_X(1) \cdot p_Y(0) = 0.60 \cdot 0.15 = 0.09$$

Conclusion: Since $0.10 \neq 0.09$, the multiplicative rule fails. Therefore, the number of smartphones (X) and smartwatches (Y) a consumer owns are statically dependent.

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Concepts of Joint Variability

When dealing with two random variables X and Y , knowing their individual means and variances is not enough. We need to measure how they **move together**.

- **Covariance:** Measures the direction of the linear relationship between two variables.
- **Correlation Coefficient:** Measures both the **direction** and the **strength** of the linear relationship on a standardized scale.

These metrics are foundational for portfolio diversification in finance, risk analysis, and regression modeling.

Covariance: Definition & Formula

The covariance of two random variables X and Y is the expected value of the product of their deviations from their respective means.

Theoretical Formula

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Computational Formula (Easier for Calculations)

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Sign Interpretation:

- $\text{Cov}(X, Y) > 0$: X and Y tend to move in the same direction.
- $\text{Cov}(X, Y) < 0$: X and Y tend to move in opposite directions.

Properties of Covariance

Let X and Y be random variables, and a, b, c, d be constants:

Fundamental Properties

- ❶ **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- ❷ **Covariance with Safeself:** $\text{Cov}(X, X) = \text{Var}(X)$
- ❸ **Constants:** $\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$
- ❹ **Variance of a Sum:**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Note on Independence: If X and Y are independent, then $\text{Cov}(X, Y) = 0$. However, the reverse is **not** always true (non-linear relationships can exist).

The Correlation Coefficient (ρ)

Covariance depends heavily on the scale of measurement (e.g., changing units from meters to centimeters changes the value). To fix this, we normalize it using standard deviations.

Pearson Correlation Coefficient Formula

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}X \cdot \text{Var}Y}}.$$

Crucial Properties

- **Boundedness:** $-1 \leq \rho \leq 1$
- $\rho = 1$: Perfect positive linear relationship.
- $\rho = -1$: Perfect negative linear relationship.
- $\rho = 0$: No **linear** relationship.

Example

Let's compute the **Covariance** and **Correlation** for our tech study example.

$X \setminus Y$	$Y = 0$	$Y = 1$	$Y = 2$	Marginal $p_X(x)$
$X = 1$	0.10	0.40	0.10	0.60
$X = 2$	0.05	0.25	0.10	0.40
Marginal $p_Y(y)$	0.15	0.65	0.20	1.00

$$EX = 1 \cdot 0.6 + 2 \cdot 0.4 = 1.4$$

$$EY = 0 \cdot 0.15 + 1 \cdot 0.65 + 2 \cdot 0.2 = 1.05.$$

Find variance X and Y :

$$\text{Var}X = EX^2 - (EX)^2 = 0.24, \quad \text{Var}Y = EY^2 - (EY)^2 = 0.3475$$

Calculate $E[XY]$

We multiply $x \cdot y \cdot p(x, y)$ for all non-zero cells in our joint table:

$$\begin{aligned} E[XY] &= (1 \cdot 1 \cdot 0.40) + (1 \cdot 2 \cdot 0.10) \\ &\quad + (2 \cdot 1 \cdot 0.25) + (2 \cdot 2 \cdot 0.10) \\ &= 0.40 + 0.20 + 0.50 + 0.40 = 1.50 \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1.50 - (1.4 \cdot 1.05) = 0.03$$

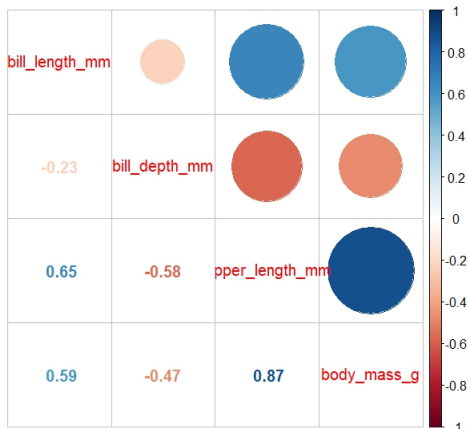
The correlation coefficient is:

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}X \cdot \text{Var}Y}} = \frac{0.03}{0.490 \cdot 0.606} \approx 0.101$$

Interpretation: A correlation of +0.101 indicates a very **weak positive linear relationship**. While X and Y are dependent (as proven earlier), their linear association is close to zero.

Covariance and Correlation in R

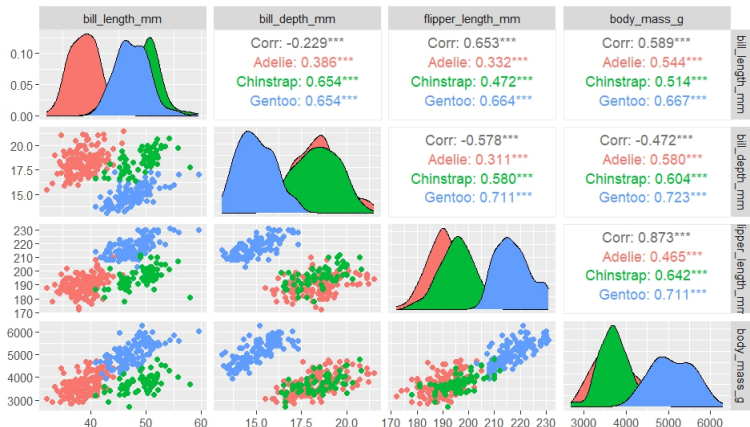
```
cov(penguins_final[,3:6])  
co<-cor(penguins_final[,3:6])  
  
library(corrplot)  
corrplot.mixed(co,lower = "number",upper = "circle")
```



Covariance and Correlation in R

```
library(GGally)

ggpairs(penguins_final, columns = 3:6, aes(color = species))
```



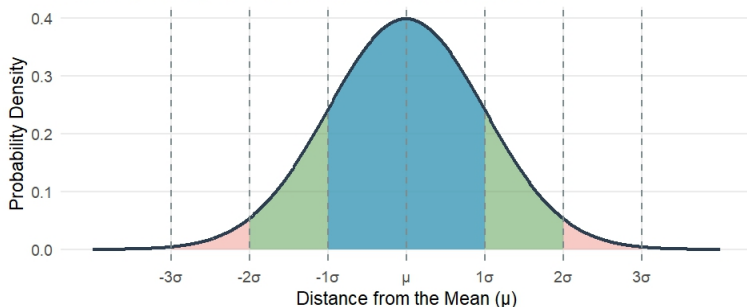
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Empirical rule for Normal Distribution

The Empirical Rule (68-95-99.7)

Probability Density Function of a Standard Normal Distribution



Data Coverage: ■ 68.3% (within 1 SD) ■ 95.4% (within 2 SD) ■ 99.7% (within 3 SD)

Markov's inequality

Theorem

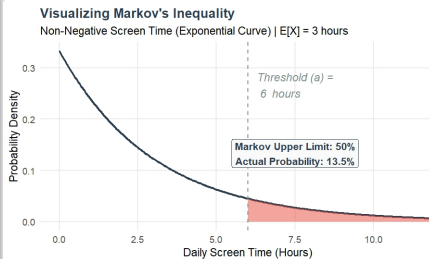
(Markov's Inequality). If X is a random variable that takes only nonnegative values, then for any value $a > 0$

$$P\{X > a\} \leq \frac{EX}{a}.$$

Daily Screen Time.

Suppose the average phone screen time for a group of teenagers is $\mu = E[X] = 3$ hours per day. We want to see the probability that a randomly selected teen uses their phone for $a = 6$ or more hours per day.

Markov's Ineq.: $\frac{3}{6} = 0.50$.



As a corollary, we obtain the following.

Theorem

(Chebyshev's Inequality). If X is a random variable with mean μ and variance σ^2 , then, for any value $k > 0$,

$$P\{|X - EX| > k\} \leq \frac{\sigma^2}{k^2}.$$

If in Chebyshev's inequality we substitute $\sigma = \sqrt{D\xi}$ then

Corollary (3σ)

If there exists a finite variance $D\xi$ then

$$P(|\xi - E\xi| \geq 3\sigma) \leq \frac{1}{9}.$$

Remark

In case of normal distributed r.v. ξ the left-hand side of inequality equals 0.003. Therefore, in engineering, it is believed that random errors during the processing do not exceed 3σ .

Example

Suppose we know that the number of items produced in a factory during a week is a random variable with mean 500.

- (a) What can be said about the probability that this week's production will be at least 1000?
- (b) If the variance of a week's production is known to equal 100, then what can be said about the probability that this week's production will be between 400 and 600?

Solution

Let X be the number of items that will be produced in a week.

(a) By Markov's inequality,

$$P\{X > 1000\} \leq \frac{EX}{1000} = \frac{500}{1000} = \frac{1}{2}.$$

(b) By Chebyshev's inequality,

$$\begin{aligned} P\{400 < X < 600\} &= P\{|X - 500| < 100\} = 1 - P\{|X - 500| \geq 100\} \\ &> 1 - \frac{100}{100^2} = \frac{99}{100}. \end{aligned}$$

so the probability that this week's production will be between 400 and 600, is at least 0.99.

If in Chebyshev's inequality we substitute $\sigma = 6\sqrt{D\xi}$ then

Corollary (6σ)

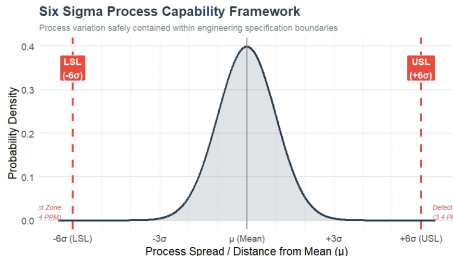
If there exists a finite variance $D\xi$ then

$$P(|\xi - E\xi| \geq 6\sigma) \leq \frac{1}{36}.$$

- **Six Sigma** is a quality management methodology based on statistics.
- Developed at Motorola (1980s), popularized by General Electric.
- The goal is to reduce the number of defects to 3.4 per million opportunities.
- Focus on efficiency, stability and customer satisfaction.

Main idea

- Control of variations (dispersion) in processes.
- Elimination of the causes of errors and defects.
- Optimization of costs and time.
- Focus on customer needs.
- Formation of a culture of continuous improvement.



DMAIC: Define – Measure – Analyze – Improve – Control

- ➊ **Define** — define the problem, goals, and customer expectations.
- ➋ **Measure** — measure current process performance.
- ➌ **Analyze** — find root causes of defects.
- ➍ **Improve** — develop solutions for improvement.
- ➎ **Control** — ensure consistency of results.

Benefits of Six Sigma

- Reduce costs and defects.
- Increase productivity and quality.
- Improve customer satisfaction.
- Transparency and measurability of processes.
- Support for the strategic development of the company.

Examples of successful implementation

- **Motorola** — increased production reliability.
- **General Electric** — savings of over \$10 billion.
- Other companies: Toyota, IBM, Samsung, Amazon.

On six sigma

The origin of six sigma

<https://www.youtube.com/watch?v=PMTEJ3I9XR0>
(1min 38 sec)

On six sigma

<https://www.youtube.com/watch?v=uB0MZ0dKQUo>
(8 min 32 sec)

Law of Large Numbers (LLN)

As the number of independent trials (n) increases, the sample average ($\bar{X}_n$) converges to the true population mean (μ).

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

- Random errors "cancel out" over time.
- This principle ensures that insurance companies and casinos remain profitable despite individual uncertainty.

Case Study: Galton's Ox (1906)

The Experiment: Francis Galton analyzed 787 guesses of a slaughtered ox's weight at a fair.

- **Actual Weight:** 1,198 lbs (approx. 543 kg)
- **Median Guess:** 1,207 lbs
- **Mean Guess:** 1,197 lbs

The Lesson

Individual errors were massive, but the "*Wisdom of the Crowd*" (the average) was accurate within 0.1%.



Рис.: Source:
www.georgeglazer.com.

The Wisdom of the Crowd & The LLN

While individual guesses fluctuated wildly (some overestimating or underestimating by hundreds of pounds), the group aggregate showed staggering accuracy.

The Law of Large Numbers (LLN)

Let $X_1, X_2, \dots, X_n$ be independent identically distributed (iid) r.v. with mean $EX_i = \mu$. Then as $n \rightarrow \infty$, the sample average converges to the true expected value:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu,$$

where convergence in probability $\xrightarrow{p}$ means

$$\forall \epsilon > 0 \quad P(|\bar{X}_n - \mu| > \epsilon) \rightarrow 0, \quad n \rightarrow \infty.$$

<https://www.youtube.com/watch?v=MntX3zWNWec&t=6s>

What is Monte Carlo Simulation?

- **Definition:** A computerized mathematical technique that allows people to account for risk in quantitative analysis and decision making.
- **The Core Idea:** Instead of solving a deterministic problem directly, we use **repeated random sampling** to obtain numerical results.
- **Origin:** Named after the Monte Carlo Casino in Monaco due to the similarity to games of chance (dice, roulette, slot machines).

Monte Carlo works because of the Law of Large Numbers (LLN).

The 3-Step Monte Carlo Process

- ❶ **Model the System:** Define the domain of possible inputs and their probability distributions.
- ❷ **Generate Inputs:** Randomly sample inputs from those distributions thousands of times.
- ❸ **Analyze Results:** Aggregate the individual deterministic outcomes to find the average, variance, and probability of specific events.

<https://huggingface.co/spaces/aranaur/lotps-explorer>

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Besides its theoretical interest and importance, this theorem provides a simple method for computing approximate probabilities for sums of independent random variables. It also explains the remarkable fact that the empirical frequencies of so many natural “populations” exhibit a bell-shaped (that is, normal) curve.

Let

$$S_n = \sum_{i=1}^n X_i.$$

Theorem

(Central Limit Theorem).

Let $X_1, X_2, \dots$ be a sequence of iid r.v. each with mean μ and variance σ^2 .

Then the distribution of

$$\frac{S_n - ES_n}{\sqrt{\text{Var}S_n}}$$

tends to the standard normal as $n \rightarrow +\infty$. That is,

$$P\left\{\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq a\right\} \approx \Phi(a),$$

where $\Phi(\cdot)$ is CDF for $N(0, 1)$.

Note that this theorem holds for any distribution of the X_i 's; herein lies its power.

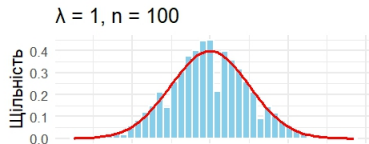
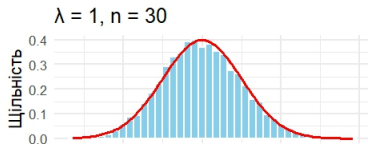
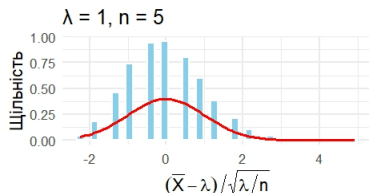
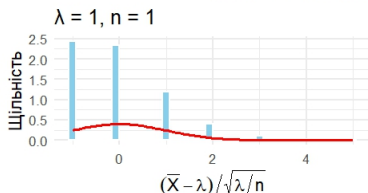
CLT in terms of $\bar{X}$

CLT for mean

If $X_1, X_2, \dots$ are iid r.v. with mean $E_i = \mu$ and variance $\text{Var}X_i = \sigma^2$ then

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{W} N(0, 1)$$

Центральна гранична теорема для Poisson($\lambda=1$)



<https://www.youtube.com/watch?v=jvoxEYmQHNM>

Example

An e-commerce distribution hub uses a cargo elevator to move outgoing shipments.

- The weights of individual packages are highly skewed and irregular, with a population mean of $\mu = 8$ kg and a standard deviation of $\sigma = 3$ kg.
- A random batch of $n = 36$ packages is loaded onto the elevator.
- The maximum safe structural weight capacity for a single elevator run is 300 kg.

The Objective

What is the probability that a random batch of 36 packages will overload the elevator (exceed 300 kg)?

Solution

Because the sample size is sufficiently large ($n = 36 \geq 30$), we can use the CLT.

The total weight of the batch is

$$S_n = \sum_{i=1}^{36} X_i$$

then

$$E[S_n] = n \cdot \mu = 36 \cdot 8 = 288 \text{ kg}$$

Standard deviation

$$\sigma_{sum} = \sqrt{n} \cdot \sigma = \sqrt{36} \cdot 3 = 6 \cdot 3 = 18 \text{ kg}$$

Therefore,

$$\begin{aligned} P(S_n > 300) &= P\left(\frac{S_n - ES_n}{\sigma_{sum}} > \frac{300 - 288}{18}\right) = \\ &= P\left(\frac{S_n - ES_n}{\sigma_{sum}} > 0.67\right) = P(Z > 0.67) = \\ &= 1 - \Phi(0.67) = 1 - 0.7486 = 0.2514. \end{aligned}$$

A total weight of 300 kg sits exactly 0.67 standard deviations above the average expected weight batch.

There is a **25.14% chance** that a random batch of 36 packages will overload the cargo elevator. Management should reduce the maximum batch size to ensure warehouse safety.

Problems

Problem 1

An organization accepts members in the top 2% of IQ scores. Assume IQ $X \sim \mathcal{N}(\mu = 100, \sigma^2 = 16^2)$.

- a. Find the minimum IQ score required for membership.
- b. Find the probability that three randomly chosen individuals from the *general population* are all admitted.
- c. Three individuals are drawn at random *from the organization*. Find the probability that all their IQs exceed 140.

*Hint: For (c), you must first find the conditional probability $P(X > 140 \mid X \geq x_{min})$.

Problems

Problem 2

The lifespan of LED light bulbs follows a normal distribution with a mean of $\mu = 10,000$ hours and a standard deviation of $\sigma = 500$ hours. What is the probability that a randomly selected bulb will last between 9,000 and 11,000 hours?

Problem 3

Consider two financial assets, X and Y . It is given that $E[X] = 5$, $E[Y] = 10$, and $E[XY] = 52$. Their standard deviations are $\sigma_X = 2$ and $\sigma_Y = 4$. Find the Pearson correlation coefficient ρ .

Problem 4

A software engineering firm tracks the number of critical server crashes per day (X) and the number of emergency hotfixes deployed by the dev team (Y) on any given day. The joint probability mass function (PMF), $p(x, y) = P(X = x, Y = y)$, is given by the table below:

$X \setminus Y$	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.40	0.10	0.00
$X = 1$	0.05	0.25	0.10
$X = 2$	0.00	0.05	0.05

- Find the marginal distributions of X and Y .
- Determine whether the number of crashes (X) and hotfixes (Y) are statistically independent.
- Calculate the Covariance $\text{Cov}(X, Y)$ and the correlation coefficient ρ_{XY} .
- Interpret your analytical results in the context of the software firm's operations.

Problems 5

The average waiting time for a customer at a clinic is $\mu = 10$ minutes, with a variance of $\sigma^2 = 4$. Find the upper bound for the probability that a customer will have to wait ≥ 20 minutes using Markov and Chebyshev inequalities.

Problem 6

A mechanical component must have a length of 50 ± 1 mm to be functional. The production line yields a process mean of $\mu = 50$ mm and a standard deviation of $\sigma = 0.2$ mm. Does this process meet the “6-Sigma” standard if not provide solution how to achieve “6-Sigma” standard?

Problems

Problem 7

A gambler flips a fair coin n times. How many times must the coin be flipped so that, with a probability of at least 0.95, the sample proportion of "Heads" deviates from 0.5 by no more than 0.01? (Use Chebyshev's Inequality).

Problem 8

A delivery truck transports 100 cargo boxes. The weight of each box is an independent random variable with a mean of $\mu = 20$ kg and a standard deviation of $\sigma = 2$ kg. What is the probability that the total cargo weight exceeds 2,050 kg?

Problem 9

An insurance company underwrites a specialized premium policy for professional digital creators against targeted cyber ransom attacks. The insurance firm has a portfolio of $n = 10,000$ independent creators. For any single creator over a 1-year timeline, the payout distribution is highly skewed: there is a 99.5% chance of zero claims, but a 0.5% chance of a catastrophic breach requiring a maximum payout of \$50,000. To fund these claims, the company charges each creator an annual premium of \$300.

- Calculate the expected value (μ) and standard deviation (σ) of a claim payout for an individual creator.
- Define the distribution of the total aggregate claims (S_n) across all 10,000 policies using the Central Limit Theorem.
- Calculate the probability that the insurance company's total annual claim payouts will exceed the total collected revenue from premiums (causing a net financial loss).

Questions?